Prime Numbers – Sieve of Eratosthenes

The Mystery Number Hook

Aarav has 7 laddoos. Can he share them equally with 2 friends? No! With 3 friends? No! This mysterious number 7 is called a PRIME NUMBER. Become a Prime Detective today!

What Is a Prime Number?

A prime number has EXACTLY 2 factors: 1 and itself. It can only be divided evenly by 1 and the number itself. Let's see some examples!

- 5 rotis = only 1×5 or 5×1
- 7 laddoos = only 1×7 or 7×1
- No other way to group them equally!

Prime vs Not Prime (Composite)

Some numbers can only be divided by 1 and themselves – these are PRIME! Others have more ways to divide – these are COMPOSITE.

- 5 beads on a maala: only 1×5 or 5×1 = PRIME!
- 6 mithai in a box: 1×6, 2×3, 3×2, 6×1 = COMPOSITE!
- 6 has MORE ways to divide = NOT prime!

The Special Number 1

Number 1 is NEITHER prime NOR composite. Remember: "Ek toh There is only one! (One is always just one!) 1 stands alone, unique and different.

Meet Eratosthenes

An ancient Greek mathematician from 2,200 years ago invented a clever method to find ALL prime numbers. He called it a SIEVE – like sieving atta (wheat flour) at home!

Lived 2,200 years ago in Greece

Invented the Sieve method to find primes

Just like Daadi sieves atta in the kitchen!

How the Sieve Works

The Sieve Analogy: Fine flour passes through (primes stay), bran gets removed (composites crossed out). We'll use this method to find primes from 1 to 30!

- Fine flour = Prime numbers that pass through
- Bran = Composite numbers that get removed
- Step-by-step process coming next!

Sieve Step 1 & 2

Let's start finding prime numbers using the Sieve of Eratosthenes! Follow these steps on our number grid from 1 to 30.

- Step 1: Cross out 1 (neither prime nor composite)
- Step 2: Circle 2 – it's our first prime number!
- Cross out all multiples of 2: 4, 6, 8, 10... up to 30

Sieve Step 3 & 4

Continue finding primes! Circle the next prime numbers and cross out their multiples from our number grid.

- Step 3: Circle 3, cross out 9, 15, 21, 27
- Step 4: Circle 5, cross out 25
- Skip numbers already crossed out!

Sieve Final Step

Step 5: Circle all remaining uncrossed numbers. These are ALL the prime numbers from 1 to 30!

- Complete prime list: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
- 10 prime numbers found!
- You completed the sieve!

Prime Number Patterns

After 2, ALL prime numbers are ODD! 2 is the ONLY even prime number – the most special one!

- 2 is the only even prime number
- All other primes are odd numbers
- Can you guess why?

Real Life Prime Examples

Prime numbers appear in everyday life! Look around and you'll find them everywhere in things we use and see daily.

- 11 rotis on a plate (can't divide equally except 1×11)
- 13 water balloons for Holi (can't share equally)
- 7 cricket stumps arrangement (only 1×7 works)

What We Learned Today

Great job, Prime Detectives! Let's remember the amazing things we discovered about prime numbers and the Sieve of Eratosthenes.

- Prime number = exactly 2 factors (1 and itself)
- Used the Sieve of Eratosthenes method
- Found all primes 1–30: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. You are now a PRIME DETECTIVE! 🔍

Fun Facts About Prime Numbers!

Did you know these amazing things about prime numbers? Math is full of surprises and wonders waiting for you to discover!

- 2 is the ONLY even prime number
- Primes with MILLIONS of digits exist!
- Aryabhata & Brahmagupta studied numbers

Thank You, Prime Detectives!

Every child deserves to love mathematics 🌟
You're a Prime Detective Champion!
Planrs Academy